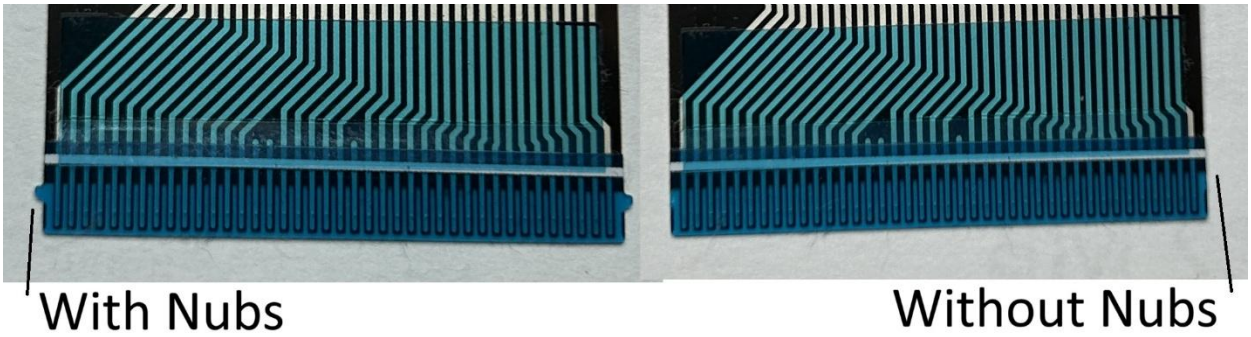
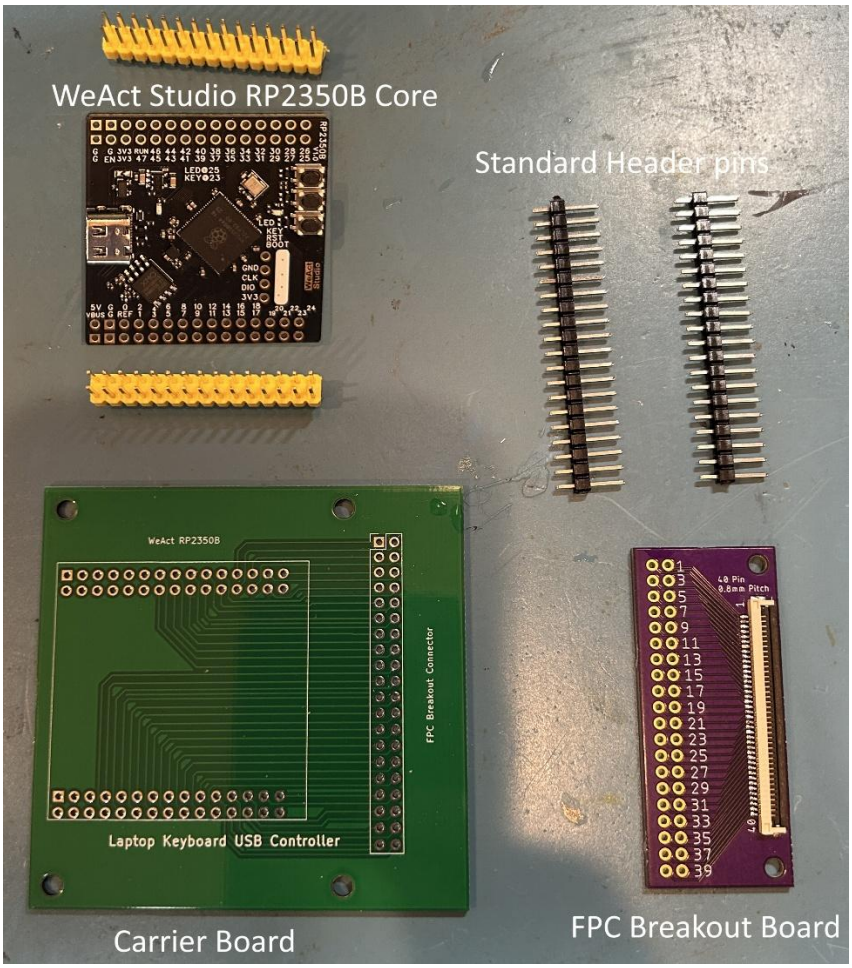
For this project, a [WeAct Studio RP2350B Core]() controls an [HP Envy X360 16 inch laptop keyboard]. The USB C connector on the WeAct module is cabled to a computer and the KMK routine scans the keys to report a key press over USB.

There were locking nubs on the cable that I trimmed off with scissors. This allowed me to use a generic FPC connector.



I designed the carrier board to connect 40 GPIO's from the WeAct Studio RP2350B Core to a standard FPC breakout board. The KiCad carrier board file and the zipped Gerber file for sending to a manufacturer like JLCPCB are at my repo.



Normally there are FPC breakout boards available from Amazon and AliExpress that fit most keyboard cables. This HP keyboard has a 40 pin FPC cable with a pitch of 0.8mm (rare) so I had to design my own FPC Breakout board and solder the connector. The breakout board Eagle files are at my repo. The FPC connector is from AliExpress.

For keyboards with less pins, install the FPC breakout board wherever is convenient. The software will determine which keyboard keys are attached to the GPIO pins. Note that GPIO's 0, 23, 25, 43-47 are not used by the carrier board.

The following table gives the FPC to GPIO pin connections. All of the software uses GPIO numbers so you should not need to translate to FPC pin numbers unless you are debugging a solder short.

FPC Pin Numbers	WeAct RP2350B GPIO
1	32
2	31
3	30
4	29
5	26
6	28
7	27
8	34
9	33
10	36
11	35
12	38
13	37
14	40
15	39
16	42
17	41
18	1
19	2
20	3
21	4
22	5
23	6
24	7
25	8
26	9
27	10
28	11
29	12
30	13
31	14
32	15
33	24
34	22
35	21
36	20
37	19
38	18
39	17
40	16

This project uses similar software and the same basic procedure for loading the libraries as described in my Hackster.IO project and step 16 of my Instructable. The WeAct Studio RP2350B will not erase the flash memory when loaded with the flash_nuke.uf2 program. I suspect I need to find a newer flash_nuke but have not investigated further. The WeAct Studio RP2350B Core was recently added to the Adafruit Circuit Python page so to get the Python uf2, use the latest file from here. To load uf2 files, hold the boot button, click the reset button, release the boot button, load the uf2 into the folder, then click reset again. I created a new version of my matrix decoder Python code to account for the additional GPIO's. It must be renamed to code.py when loading it on the WeAct RP2350B Core. It is a little squirrely to get to run, probably just my

Python and WeAct RP2350B inexperience, (if you find a bug, let me know). I'll keep working on improving the code. The GPIO connection list for all the HP Envy X360 keys is given below.

KC.LCTRL	24	22
KC.RCTRL	24	21
KC.LSHIFT	20	18
KC.RSHIFT	18	16
KC.LALT	19	5
KC.RALT	13	5
KC.LGUI	17	7
FN	14	8
KC.A	21	12
KC.B	13	11
KC.C	20	10
KC.D	21	10
KC.E	16	10
KC.F	21	11
KC.G	17	11
KC.H	17	9
KC.I	16	6
KC.J	21	9
KC.K	21	6
KC.L	21	3
KC.M	20	9
KC.N	13	9
KC.O	16	3
KC.P	41	16
KC.Q	16	12
KC.R	16	11
KC.S	21	15
KC.T	22	11
KC.U	16	9
KC.V	20	11
KC.W	16	15
KC.X	20	15

KC.Y	22	9
KC.Z	20	12
KC.GRAVE	14	12
KC.N1	19	12
KC.N2	19	15
KC.N3	19	10
KC.N4	19	11
KC.N5	14	11
KC.N6	14	9
KC.N7	19	9
KC.N8	19	6
KC.N9	19	3
KC.N0	41	19
KC.MINUS	41	14
KC.EQUAL	41	13
KC.BSPACE	17	2
KC.ESC	17	12
KC.F1	13	10
KC.F2	14	10
KC.F3	22	10
KC.F4	17	10
KC.F5	15	14
KC.F6	22	15
KC.F7	17	15
KC.F8	15	13
KC.F9	14	3
KC.F10	22	3
KC.F11	17	3
KC.F12	13	3
KC.DELETE	13	1
KC.RIGHT	19	1
KC.LEFT	16	1
KC.UP	16	2
KC.DOWN	19	2
KC.SLASH	41	20

KC.DOT		20	3
KC.COMMA		20	6
KC.SCOLON		41	21
KC.QUOTE		41	17
KC.ENTER		21	2
KC.LBRC		41	22
KC.RBRC		22	6
KC.BSLASH		14	2
KC.CAPS		13	12
KC.TAB		22	12
KC.SPACE		20	2
KC.HOME		22	1
KC.END		20	1
KC.PGUP		14	1
KC.PGDOWN		21	1
KC.AUDIO_MUTE	Func	15	14
KC.AUDIO_VOL_UP	Func	17	15
KC.AUDIO_VOL_DOWN	Func	22	15
KC.BRIGHTNESS_UP	Func	22	10
KC.BRIGHTNESS_DOWN	Func	14	10
KC.MEDIA_NEXT_TRACK	Func	17	3
KC.MEDIA_PREV_TRACK	Func	14	3

The Normal key matrix for the HP Envy X360 16 inch keyboard.

GPIO#	13	14	16	17	19	20	21	22
1	DEL	PGUP	LEFT		RIGHT	END	PGDOWN	HOME
2		BSLASH	UP	BSPACE	DOWN	SPACE	ENTER	
3	F12	F9	O	F11	N9	DOT	L	F10
5	RALT				LALT			
6			I		N8	COMMA	K	RBRC
7				LGUI				
8		FN						
9	N	N6	U	H	N7	M	J	Y
10	F1	F2	E	F4	N3	C	D	F3
11	B	N5	R	G	N4	V	F	T
12	CAPS	GRAVE	Q	ESC	N1	Z	A	TAB
15	F8	F5	W	F7	N2	X	S	F6
18			RSHIFT			LSHIFT		
24							RCTRL	LCTRL
41	EQUAL	MINUS	P	QUOTE	N0	SLASH	SCOLON	LBRC

The Media matrix is the same as above except with the desired Fn-Media keys (shown in bold).

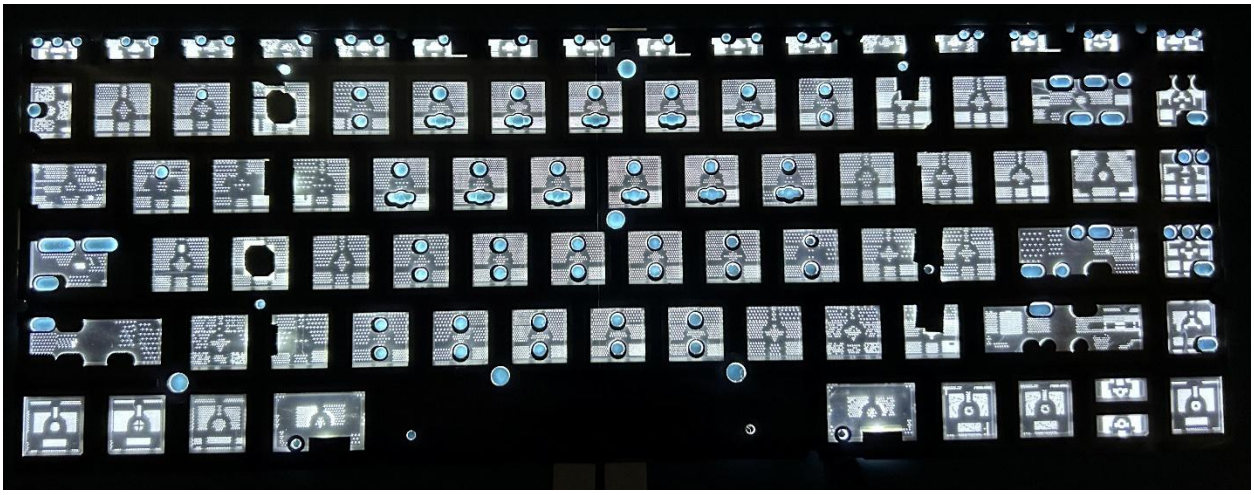
	13	14	16	17	19	20	21	22
1	DEL	PGUP	LEFT		RIGHT	END	PGDOWN	HOME
2		BSLASH	UP	BSPACE	DOWN	SPACE	ENTER	
3	F12	**MPRV**	O	**MNXT**	N9	DOT	L	F10
5	RALT				LALT			
6			I		N8	COMMA	K	RBRC
7				LGUI				
8		FN						
9	N	N6	U	H	N7	M	J	Y
10	F1	BRID	E	F4	N3	C	D	**BRIU**
11	B	N5	R	G	N4	V	F	T
12	CAPS	GRAVE	Q	ESC	N1	Z	A	TAB
15	F8	**MUTE**	W	**VOLU**	N2	X	S	**VOLD**
18			RSHIFT			LSHIFT		
24							RCTRL	LCTRL
41	EQUAL	MINUS	P	QUOTE	N0	SLASH	SCOLON	LBRC

I modified Marcel's Python code to convert the connection list into a normal key matrix. The code runs in Thonny on a PC and uses this keyboard pin list text file. I also made the matrix manually to check the program and they were the same. The matrixes were put into KMK and worked fine with no odd behaviors. The KMK code_Envy360.py must be renamed to code.py when loading into the WeAct RP2350B Core.

Keyboard Backlight – The keyboard came with a separate backlight so it can be applied to the back of the keyboard by pulling off the tabs and exposing the adhesive.

The backlight consists of a simple set of diodes all wired in parallel and connected to a 4 pin 1mm pitch FPC cable. Setting my ohm meter to diode mode showed 2 pins are tied together for the anode and 2 pins are tied together for the cathode. Normally a PWM signal is used to control a FET that can handle the current of the backlight diodes. I believe KMK can be programmed to control a backlight but for a simple test, I just used my bench power supply. At 2.5 volts, the backlight was just barely visible and draws 7ma. The current increased exponentially to 150ma at 3.3 volts with full brightness.

This picture shows the illuminated backlight without the keyboard.



This picture shows the illuminated backlight under the keyboard.

Conclusion - As the WeAct module gets more users and the libraries and documentation become more mature, it should be a popular way to make a USB laptop keyboard controller thanks to its low price and high I/O count.